

Aliaksei Krents

Sr. Full-Stack Software Engineer | Python, Django, FastAPI, React, AWS, AI/ML



Summary

Senior Full-Stack Software Engineer with 8+ years of experience building AI-powered SaaS, real-time communication, healthcare, analytics, and enterprise web/mobile platforms. Strong expertise in Python, TypeScript, Django, FastAPI, React, Next.js, React Native, PostgreSQL, Redis, AWS, Docker, and Kubernetes, with hands-on experience designing distributed systems, secure APIs, microservices, WebRTC workflows, and scalable data pipelines. Skilled in integrating GenAI, OpenAI APIs, LangChain, RAG, Speech-to-Text, Text-to-Speech, vector search, and automation workflows to deliver intelligent, high-performance products. Proven ability to lead end-to-end engineering from architecture and backend development to frontend delivery, cloud deployment, CI/CD, testing, observability, and production optimization.



Work Experience

2024.4
2026.4

Senior Full-Stack Software Engineer

SoftKraft | Germany

- Architected a real-time multilingual meeting platform using **Python, TypeScript, Django REST Framework, FastAPI, WebRTC**, and **REST APIs**, supporting instant calls, scheduled meetings, QR-based joining, participant control, and live translated communication across web and mobile clients.
- Designed scalable backend microservices with **DDD, gRPC, GraphQL, WebSocket**, and **Third-Party API Integration**, enabling 1:1 calls, group sessions, speaker/listener roles, moderation workflows, and conference sessions for thousands of listeners.
- Built a real-time AI speech intelligence pipeline, **Speech → Streaming STT → Translation → LLM Processing → TTS → Captions**, using **OpenAI APIs (GPT-4o), Whisper-style ASR, Azure Cognitive Services, Azure Neural TTS, PyTorch, LangChain, LlamaIndex, RAG, Embeddings, Vector Search, pgvector**, converting live speaker audio into translated captions, contextual summaries, and synthesized speech with sub-second perceived latency.
- Optimized high-volume data workflows with **PostgreSQL, MySQL, Redis, AWS ElastiCache, Elasticsearch, AWS OpenSearch**, indexing, and partitioning, improving transcript retrieval, call history access, session search, and analytics performance for massive meeting datasets.
- Implemented secure communication and access control using **JWT, OAuth 2.0, OIDC, RBAC, Cloudflare WAF**, and Rate Limiting, protecting meeting rooms, admin operations, QR links, map discovery, and public conference join flows.
- Deployed cloud-native services with **Docker, Kubernetes, AWS (EKS, ECR, RDS, S3, CloudFront, Route 53, IAM, Secrets Manager, CloudWatch)**, and **AWS EC2-based RTC/media infrastructure**, and **CI/CD** pipelines, supporting horizontal scaling of RTC, and TTS workloads.
- Built polished, accessible, and SEO-friendly client experiences with **React, Next.js, React Native, Storybook**, React Query, Redux Toolkit, Refine, SSR, SSG, WCAG 2.2, SEO, and **UI/UX** best practices, while ensuring engineering quality with TDD, GitHub Actions, pull requests, code reviews, **Agile/Scrum**, Cursor, and PyCharm.

2021.12
2024.3

Senior Software Engineer

Monterail | Wrocław, Poland

- Led backend engineering using **Python, Django, AWS EC2, WebRTC (LiveKit)** to deliver real-time multilingual conferencing with AI transcription, reducing manual documentation by 70% and improving collaboration efficiency.
- Architected distributed systems with **FastAPI, gRPC, RESTful APIs, DDD, OOP**, enabling low-latency communication pipelines and stable performance across 50+ concurrent sessions.
- Designed high-performance data and search layers using **RDBMS, PostgreSQL, SQLAlchemy, query optimization**, enabling millisecond-level transcript retrieval and analytics.
- Built asynchronous processing systems with **Redis, Celery**, pub/sub messaging, task orchestration, supporting real-time AI inference and non-blocking workloads.
- Engineered cloud-native infrastructure using **Docker, Kubernetes (EKS)**, AWS ECR, CloudWatch, CI/CD pipelines, GitHub Actions, ensuring scalability, observability, and resilient production environments.
- Integrated GenAI capabilities using **OpenAI APIs (GPT-4/4o), LangChain, RAG, Claude, Copilot**, Speech-to-Text, enabling real-time summarization and multilingual intelligence.
- Developed front-end features using **React, TypeScript, D3.js**, and WCAG standards while implementing advanced SEO with **Next.js SSR/SSG**, dynamic rendering, React Helmet, Core Web Vitals (LCP, CLS, INP, FCP, TTFB), schema markup, XML sitemap, robots.txt, canonical tags, CDN caching, and image optimization.



Personal Info



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Education

B.S. in Computer Science

Belarusian Trade and Economics
University of Consumer Cooperatives
(BTEU)

2013.9 - 2017.8



Competence

- AI-Powered Full-Stack Engineering
- SaaS Platform Development
- Real-Time Communication Systems
- Healthcare Technology Solutions
- Web & Mobile Application Development
- Cloud-Native Architecture
- Distributed Systems & Microservices
- API Design & Third-Party Integration
- GenAI, RAG & LLM Workflows
- WebRTC, Speech-to-Text & Text-to-Speech
- Database Optimization & Data Pipelines
- CI/CD, Testing & Production Reliability



Work Experience

2019.5
2021.11

Software Engineer

Vilmate LLC | Warszawa, Poland

- Built an AI voice assistant for memory care using **Python, TypeScript, FastAPI, Conversational AI, and NLP**, helping patients ask simple questions about routines, location, family, and past events through calm voice-based support.
- Developed patient and caregiver workflows with **Django, PostgreSQL, REST APIs, and healthcare data modeling**, supporting family-connected access across assisted living, homecare, and virtual care environments.
- Implemented contextual memory retrieval using **Retrieval Pipelines, Memory Graph, personal context processing, and response tuning**, transforming notes, photos, routines, and family history into structured recall support.
- Created voice interaction flows with **Speech-to-Text, Text-to-Speech, Voice Personalization, emotional intent detection, and calm response generation**, enabling familiar voice experiences for repetitive daily questions.
- Designed secure healthcare data handling with **encrypted storage, audit logging, role-based permissions, protected voice data workflows, and secure cloud deployment**, protecting patient memories, routines, and caregiver-only information.
- Delivered accessible care interfaces using **React, React Native, TypeScript, caregiver dashboards, and simple UI/UX flows**, improving daily usability for patients, families, and care teams while reducing repetitive caregiver explanations by up to **25 to 35%** and improving routine adherence by **20 to 30%**.

2017.9
2019.4

Full Stack Engineer

Euristiq | Warszawa, Poland

- Engineered a high-performance impact analytics backend using **Python, FastAPI, and AWS**, delivering **4x faster biomechanical data processing** for athlete injury research.
- Designed distributed data orchestration services with **GraphQL, Apollo Client, and Redis**, unifying multi-sensor and video streams into real-time research endpoints.
- Built immersive 3D visualization dashboards using **TypeScript, Next.js, and Three.js**, transforming raw telemetry into synchronized playback, violin plots, and 3D heatmaps.
- Implemented secure research data architecture with **PostgreSQL, JWT, and RBAC**, enforcing structured schema validation and multi-tier access governance.
- Integrated AI workflows using **TensorFlow, OpenAI API, and GitHub Copilot**, enabling automated impact classification and faster feature development.
- Strengthened platform quality through **Jest, React Testing Library, and Pytest**, ensuring cross-layer reliability across continuous Scrum releases.
- Delivered scalable cloud infrastructure using **Docker, Kubernetes, and GitHub Actions**, establishing CI/CD pipelines to support white-label product expansion.



Projects

AI-Powered Enterprise Video Conferencing Platform for Real-Time Transcription, Translation, and Automated Meeting Intelligence

<https://svensacall.com/>

SVENSA Voice is a secure AI-powered enterprise communication platform built using Python, Django, FastAPI, Django REST APIs, MySQL, LiveKit, LangChain, Python microservices, and AWS EKS, unifying video conferencing, live transcription, real-time voice translation, smart scheduling, recordings, and AI-generated summaries in one workspace. Designed for global teams, healthcare communication, telemedicine, and multilingual collaboration, it supports up to 50 meeting participants, unlimited live stream viewers, secure join links, screen sharing, admin controls, and configurable retention policies. Leveraging scalable backend microservices, real-time communication infrastructure, dedicated AI speech processing, and secure cloud storage, the platform significantly reduced manual documentation effort by up to 70% while ensuring high availability and low-latency performance.



Skills

Programming Languages:

Python, TypeScript, JavaScript, SQL, HTML5, CSS3, Bash

Backend Engineering:

Django, Django REST Framework, FastAPI, Node.js, RESTful APIs, GraphQL, gRPC, WebSocket, Microservices,DDD, OOP, API Architecture, Background Processing

Frontend Engineering:

React, Next.js, React Native, Redux Toolkit, React Query, Refine, Storybook, D3.js, Three.js, SSR, SSG, Tailwind CSS, MUI, Bootstrap, Responsive UI, WCAG

AI & LLM Engineering:

OpenAI APIs, GPT-4o, Claude, LangChain, LlamaIndex, RAG, Prompt Engineering, Embeddings, Vector Search, NLP, AI Transcription, AI Summarization

Voice & Real-Time AI:

Speech-to-Text, Text-to-Speech, Live Translation, Real-Time Captions, Conversational AI, WebRTC, LiveKit, Audio Streaming, RTC Media Infrastructure

Databases & Search:

PostgreSQL, MySQL, Redis, SQLAlchemy, pgvector, Elasticsearch, AWS OpenSearch, Data Modeling, Indexing, Query Optimization, Caching

Cloud & Infrastructure:

AWS EC2, EKS, ECR, RDS, S3, CloudFront, Route 53, IAM, Secrets Manager, CloudWatch, GCP, Azure, ElastiCache, Docker, Kubernetes, Cloudflare

DevOps & Reliability:

GitHub Actions, Jenkins, CI/CD, Containerized Deployment, Automated Releases, Production Monitoring, Observability, Performance Optimization, Troubleshooting

Security & Access Control:

JWT, OAuth 2.0, OIDC, RBAC, Passwordless Magic Link, Cloudflare WAF, Rate Limiting, Encrypted Storage, Audit Logging, Secure API Design

Testing & Collaboration:

TDD, Pytest, Unittest, Jest, React Testing Library, Playwright, Code Reviews, Pull Requests, Agile/Scrum, Jira, Git, GitHub, Cursor, PyCharm